DYNAMICAL SYSTEMS
PROBLEM SET 3

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We begin by noting that all of the files referenced in this document are available at:
https://github.com/chtompki/MathFall2026/tree/main/DynamicalSystems/ProblemSet3

Chapter 6

6.2.2 Consider the system

$$\dot{x} = y, \qquad \dot{y} = -x + (1 - x^2 - y^2)y.$$

a) Let D be the open disk $x^2 + y^2 < 4$. Verify that the system satisfies the hypotheses of the existence and uniqueness theorem throughout the domain D.
b) By substitution, show that $x(t) = \sin t$, $y(t) = \cos t$ is an exact solution of the system.
c) Now consider a different solution, in this case starting from the initial condition $x(0) = \frac{1}{2}$, $y(0) = 0$. Without doing any calculations, explain why this solution must satisfy $x(t)^2 + y(t)^2 < 1$ for all $t < \infty$.

6.5.20 (Rock-paper-scissors) In the children's hand game of rock-paper-scissors, rock beats scissors (by smashing it); scissors beats paper (by cutting it); and paper beats rock (by covering it). In a biological setting, analogs of this non-transitive competition occur among certain types of bacteria (Kirkup and Riley 2004) and lizards (Sinervo and Lively 1996).

Consider the following idealized model for three competing species locked in a life-and-death game of rock-paper-scissors:

$$\dot{P} = P(R - S), \qquad \dot{R} = R(S - P), \qquad \dot{S} = S(P - R),$$

where P, R, and S (all positive, of course) are the sizes of the paper, rock, and scissors populations.

a) Write a few sentences explaining the various terms in these equations. Be sure to comment on why a given term has a plus or minus sign in front of it. You don't have to write much—just enough to demonstrate that you understand how the form of the equations reflects the rock-paper-scissors story. Also, state some of the biological assumptions being made here implicitly.
b) Show that $P + R + S$ is a conserved quantity.
c) Show that PRS is also conserved.
d) How does the system behave as $t \to \infty$? Prove that your answer is correct. (Hint: Visualize the level sets of the functions $E_1(P, R, S) = P + R + S$ and $E_2(P, R, S) = PRS$ in the three-dimensional (P, R, S) space. What can you infer from the fact that all trajectories must simultaneously lie in level sets of both functions?)

6.6.4 (Computer explorations) For each of the following reversible systems, try to sketch the phase portrait by hand. Then use a computer to check your sketch. If the computer reveals patterns you hadn't anticipated, try to explain them.

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a) $\ddot{x} + \dot{x}^2 + x = 3$
b) $\dot{x} = y - y^3, \qquad \dot{y} = x\cos y$
c) $\dot{x} = \sin y, \qquad \dot{y} = y^2 - x$

6.7.3 (Nonlinear damping) Analyze

$$\ddot{\theta} + (1 + a\cos\theta)\dot{\theta} + \sin\theta = 0,$$

for all $a \geq 0$.

6.8.7 (Ruling out closed orbits) Use index theory to show that the system

$$\dot{x} = x(4 - y - x^2), \qquad \dot{y} = y(x - 1)$$

has no closed orbits.

Chapter 7

7.2.9 For each of the following systems, decide whether it is a gradient system. If so, find V and sketch the phase portrait. On a separate graph, sketch the equipotentials $V =$ constant. (If the system is not a gradient system, go on to the next question.)

a) $\dot{x} = y + x^2 y, \qquad \dot{y} = -x + 2xy$
b) $\dot{x} = 2x, \qquad \dot{y} = 8y$
c) $\dot{x} = -2xe^{x^2+y^2}, \qquad \dot{y} = -2ye^{x^2+y^2}$

7.2.12 Show that

$$\dot{x} = -x + 2y^3 - 2y^4, \qquad \dot{y} = -x - y + xy$$

has no periodic solutions. (Hint: Choose a, m, and n such that $V = x^m + ay^n$ is a Liapunov function.)

7.2.18 Consider the predator-prey model

$$\dot{x} = rx\left(1 - \frac{x}{2}\right) - \frac{2x}{1+x}y, \qquad \dot{y} = -y + \frac{2x}{1+x}y,$$

where $r > 0$ and $x, y \geq 0$. Prove this system has no closed orbits by invoking Dulac's criterion with the function

$$g(x, y) = \frac{1+x}{x}\, y^{\alpha - 1}$$

for a suitable choice of α (Hofbauer and Sigmund 1998).

7.3.1 Consider

$$\dot{x} = x - y - x(x^2 + 5y^2), \qquad \dot{y} = x + y - y(x^2 + y^2).$$

a) Classify the fixed point at the origin.
b) Rewrite the system in polar coordinates, using $r\dot{r} = x\dot{x} + y\dot{y}$ and $\dot{\theta} = (x\dot{y} - y\dot{x})/r^2$.
c) Determine the circle of maximum radius, r_1, centered on the origin such that all trajectories have a radially outward component on it.
d) Determine the circle of minimum radius, r_2, centered on the origin such that all trajectories have a radially inward component on it.
e) Prove that the system has a limit cycle somewhere in the trapping region $r_1 \leq r \leq r_2$.

7.5.7 (Cell cycle) Tyson (1991) proposed an elegant model of the cell division cycle, based on interactions between the proteins cdc2 and cyclin. He showed that the model's mathematical essence is contained in the following set of dimensionless equations:

$$\dot{u} = b(v-u)(B+u^2) - u, \qquad \dot{v} = c - u,$$

where u is proportional to the concentration of the active form of a cdc2-cyclin complex, and v is proportional to the total cyclin concentration (monomers and dimers). The parameters $b \gg 1$ and $B \ll 1$ are fixed and satisfy $8Bb \ll 1$, and c is adjustable.

a) Sketch the nullclines.

b) Show that the system exhibits relaxation oscillations for $c_1 < c < c_2$, where c_1 and c_2 are to be determined approximately. (It is too hard to find c_1 and c_2 exactly, but a good approximation can be achieved if you assume $8Bb \ll 1$.)

c) Show that the system is excitable if c is slightly less than c_1.